



Advanced Technological Institute – Kurunegala

Sri Lanka Institute of Advanced Technological Education

HNDIT 2402 -Data Communication and Networks

Higher National Diploma Information Technology

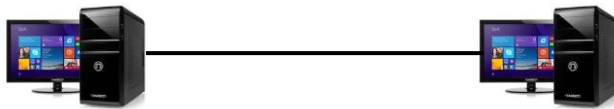
Lab sheet 2-Part II

Design Simple Network (Introduction to Packet Racer)

In this lab session, you will hands on experience in Packet Racer Software. Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate modern computer networks.

Part 1

1. Design a following Network using packet racer software



2. Assign suitable IP addresses and check connectivity. Briefly explain your steps.

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3. Modify above network to connect a switch or a hub and check the connectivity.

4. What are the cable types use in above two networks?

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5. Modify above network to connect wireless PC or Laptop. Check the connectivity.

6. What are the additional device you need to establish above network?

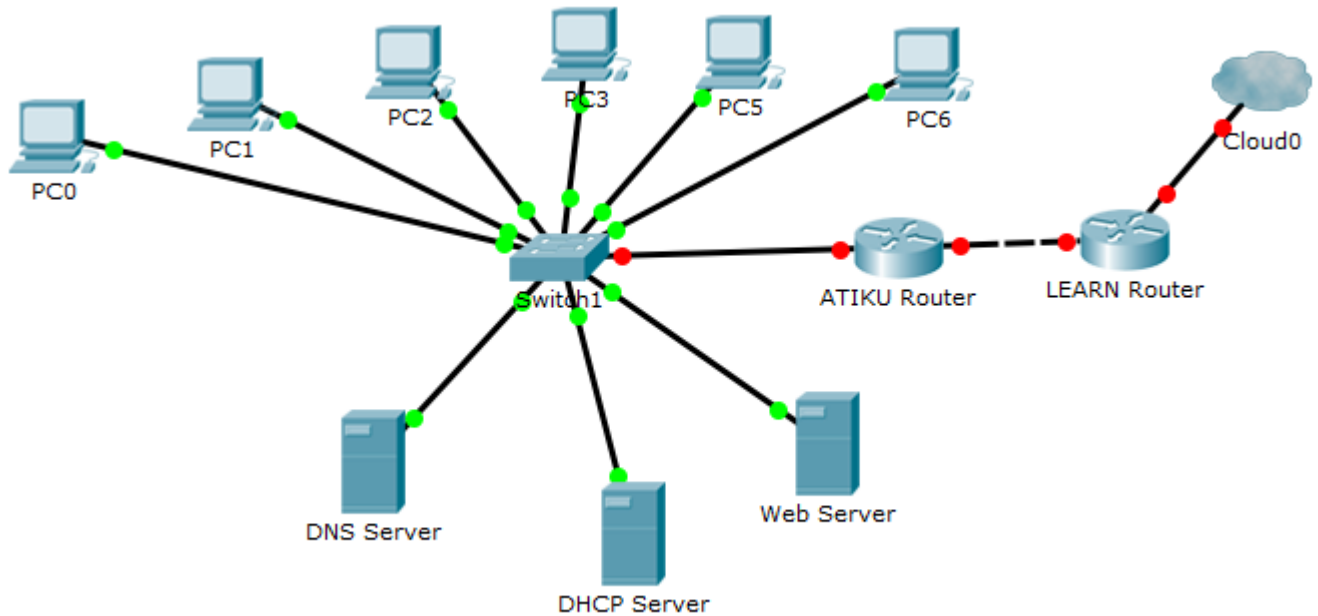
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7. Briefly explain your steps followed in above network design.

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Part 2

1. Design the following network using packet racer software.



ATIKU-Network

2. Give static IPs to Web Server, DHCP Server, DNS Server. Write down given IP addresses.

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3. Configure DHCP Server. Write your Steps.

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4. Check whether PCs take IPs. Write IP's of PCs.

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5. Check the connectivity between PCs and Servers.

6. Configure the web server. Create a simple web site there. Briefly explain your step.

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7. Visit your web site using a PC (using IP address). Briefly explain the step.

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8. Configure DNS server. Add DNS records for your previously created site (atiku.ac.lk). Write the steps you have done there.

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9. Visit your website using domain name.

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10. Modify above network to establish more web servers/web sites. Add DNS records for above web servers/websites. Try to access above sites using PCs. Write the steps you have done there

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11. How can you give access your ATI website to outside world? Your internet provider is LEARN (Lanka Education and Research Network). Website name: atiku.ac.lk

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12. If you are going to use website name as **atiku.com**, how can give access that site to outside world?

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13. Suggest a way to connect two networks which are physically nearby.

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14. Give your suggestion way to connect our ATI network and Jaffna ATI network.

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